

Quiet and Efficient Vent Fans

According to the Home Ventilating Institute (HVI), the humidity level in a bathroom during a bath or shower “is like that in a tropical rain forest—uncomfortable, hot, and damaging—a perfect breeding ground for mold, mildew, and micro-organisms that can impact your health.” Vent fans, whether used in the bathroom, utility room, or kitchen, play a key role in maintaining appropriate humidity levels and removing objectionable odors. While HVI recommends that bathroom exhaust fans run for 20 minutes after each bathroom use, these fans are often not used as intended either because they are too noisy or, less frequently, because they use too much energy. As a result, homeowners suffer from poor air quality and potential health risks, and contractors receive complaints.

The good news is that the options for quiet and efficient vent fans are growing. There are now over 150 qualified models on the EPA’s Energy Star product list (see “Energy Star Specification”). These include bathroom/utility vent fans, inline fans, and kitchen range hoods. A key feature of these qualified models is noticeably quieter operation, making it much more likely that homeowners will use them. In fact, an efficient, high-quality, vent fan can be a selling point to a potential home buyer. More and more contractors and homeowners are recognizing the benefits of these products, and manufacturers are making more models accessible to consumers.

Vent Fan as Selling Point

A rule of thumb in real estate is that kitchens and bathrooms receive the most scrutiny from potential home buyers. An otherwise positive first impression of a home can be ruined at the flip of a switch—if a loud vent fan comes roaring to life. Rather than walking potential buyers through a house



Reversomatic’s 4000-250 ES Range Hood is the first Energy Star-qualified model.

Table 1. Average Power Requirements for Bathroom Vent Fans with Lighting (80–150 CFM)

	Power (Watts)		
	Fan	Light	Total
Standard-Efficiency	59	100	159
Energy Star	33	26	59
Savings	26 (45%)	174 (74%)	100 (63%)

while fearing that they will switch on a noisy bath fan, builders can show the bathroom and point to the quiet fan as an added benefit. When home buyers learn that the fans will also save them money through reduced electricity bills, they are further impressed (see Table 1).

One of the more notable success stories from contractors using Energy Star-qualified fans comes from Walter Rodriguez of Master Cooling Corpora-

tion. “The fans can even be a selling point for builders,” says Rodriguez. Prior to his most recent project, phase 2 of the Arverne by the Sea townhouse community in Queens, New York, Rodriguez had used only standard-efficiency models. Since the homes at Arverne were designed to be high-performance houses, with characteristically tighter building envelopes, Rodriguez

wanted to be sure that they would receive adequate ventilation and so placed the fans in some bathrooms on automatic timers to run periodically throughout the day. Knowing that homeowners would not tolerate standard, noisy fans, Rodriguez looked for alternatives and found Panasonic’s Whisper Series line of fans. In addition to offering ultra-low sone levels (0.4 and 0.8 sones for the models selected for



This remodeled bathroom features a Fantech Energy Star inline fan.

www.homeenergy.org/links.html

Arverne, compared to an industry average of over 2.5 sones), the fans also offered energy savings of up to 65%.

After testing a few of the fans, Rodriguez was convinced. “I was really

shocked. It really is a better product. The 8-inch Panasonic fan is quieter than a 4-inch [standard] fan.” He decided to specify this model for all phase 2 bathroom and kitchen fans—not just those on

timers. With six vent fans per unit in all 1,200 homes, each homeowner in Arverne will save about \$19 per year, for a communitywide annual savings of approximately \$23,000.

Energy Star Specification

In order to qualify for Energy Star, a ventilation fan must pass a test administered by the HVI. The test measures the fan air flow in CFM and the power draw at a specific static pressure, which varies with the type of fan being tested. Single-port fans (those with only one ducted inlet/outlet), such as bathroom or utility room fans, are tested at 0.1 inches WC (about 250 Pa) static pressure. Fans with two or more ports are measured at 0.2 inches WC static pressure.

Fans must demonstrate their efficacy—the volume of air moved per minute for each watt of power, measured in CFM/W. Products with efficacy values greater than or equal to the standard meet this second requirement for quali-

Table. Energy Star Ventilation Fan Performance Criteria

Ventilation Fan Category	Max Air flow (CFM)	Minimum Efficacy (CFM/W)	Max Noise Level (Sones)
Kitchen range hood	500	2.8	2
Bathroom/utility room	80	1.4	2
Bathroom/utility room	500	2.8	2/3
Inline (single- and multiport)	None Specified	2.8	None Specified

cation. In addition to demonstrating energy efficiency, Energy Star-qualified ventilation fans must also demonstrate quiet operation. The HVI tests measure noise using the linear sone scale. For reference, 1 sone is equivalent to the noise of a very quiet refrigerator, while 2 sones

fans and range hoods that have lighting must also have efficient light sources. They must not accept incandescent light-bulbs, and they must meet certain other performance criteria to ensure long-lasting, aesthetically pleasing light.

(twice as loud as 1 sone) might describe the sound of quiet conversation. To meet the noise specification, vent fans must not exceed the posted level (see Table). Furthermore, vent



BROAN-NU-TONE

Select Broan-NuTone QT Series bath fans are now available through Lowe's and The Home Depot.

Sylvain Côté of Absolute Remodeling has found that Energy Star-qualified inline fans produced by Fantech provide energy savings as well as the traditional benefits of inline fans. Traditionally used in higher-end projects, inline fans provide extra flexibility and performance. Because the fan is remotely mounted, often in the attic, noise from inline fans is often muffled by the ductwork, so they tend to be less noisy than standard fans. In addition, multiport models allow multiple rooms to be ventilated by a single motor. Using Fantech's qualified models in nearly all of his kitchen and bath projects, Côté strives to provide his clients with "the best product on the market." How do his customers react to their new exhaust fan? "They all love it. Everybody is used to the noisy fans we have had to endure. These are just so much different." Energy Star-qualified inline fans cost about the same as standard-efficiency alternatives, but they use about 25% less energy.

Retail Availability Growing

One of the biggest names in the market, Broan-NuTone has 85 models on the Energy Star-qualified list. Models in their QT series are now available at Lowe's and The Home Depot,

among other retail outlets. As they got ready to introduce their QT series at these stores, Broan-NuTone modified packaging and display bays to prominently identify the qualified products.

Advanced Lighting Package

Energy Star kitchen range hoods and bathroom exhaust fans with a light are eligible fixtures for the Energy Star Advanced Lighting Package (ALP). The ALP is a new-construction option that upgrades roughly half the fixtures to Energy Star-qualified models. In addition to saving the homeowner money by using two-thirds less energy than standard fixtures, qualified fixtures provide the convenience of long-lasting bulbs that only need to be changed only once every seven years on average. Some utilities throughout the country also offer incentives for homeowners who select the package.

More information about the ALP can be found at www.energystar.gov/index.cfm?c=fixtures.alp_consumers.

In place of a 100W incandescent bulb, the models with integrated lighting use a 26W pin-based CFL. (For more on Energy Star fans with light fixtures, see "Advanced Lighting Package.") These models currently employ magnetic ballasts, but Broan-NuTone is in the process of converting all of its Energy Star-qualified models to include electronic ballasts. Utilities in the states of Massachusetts, Rhode Island, and Connecticut are treating these fans as qualifying light fixtures for Interior Light Fixture incentives. Vent fans should be added to the next round of instant coupons (worth \$10–\$15, depending on the jurisdiction) delivered to stores.

Energy Star Range Hood Makes Its Debut

The newest addition to the Energy Star family of ventilation products is the first qualified kitchen range hood. The Deluxe 4000-250ES range hood by Reversomatic provides an efficacy of 2.92 CFM/W at 140 CFM and meets the 2-sone maximum noise level limit. This constitutes a 62% reduction in power consumption and makes the 4000-2500ES range hood 70% quieter than standard models. Like many other Energy Star-qualified products, the Reversomatic range hood, at a cost of \$300–\$450 depending on the size and flow rate, is priced competitively with other midrange units.



—David Shiller

The EPA's David Shiller is the product marketing manager for Energy Star ventilation fans.

FOR MORE INFORMATION:

For more information about Energy Star ventilation products or for a complete listing of qualified products and manufacturers, visit the Energy Star Web site at www.energystar.gov. You can also contact David Shiller at the EPA: Tel: (202)343-9397; E-mail: shiller.david@epa.gov.

Or visit hvi.org.